

The Primordial Lithium Problem and FFGFT

A Brief Summary for the IPI Correspondence

Johann Pascher

ORCID: [0009-0000-6518-4064](https://orcid.org/0009-0000-6518-4064)

GitHub: <https://github.com/jpascher/T0-Time-Mass-Duality>

Zenodo (FFGFT v1.0.13): <https://zenodo.org/records/20041529>

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Abstract

The primordial lithium problem – the persistent factor 3–4 discrepancy (4.2σ – 5.3σ) between Big Bang Nucleosynthesis (BBN) predictions for ${}^7\text{Li}$ and observations of metal-poor halo stars (the Spite plateau) – has resisted resolution for four decades. Standard solution classes (astrophysical depletion, nuclear physics, beyond-Standard-Model physics) all face structural difficulties. Fields (2011) explicitly lists “large changes to the cosmological framework” as a possible solution class. FFGFT instantiates that class: in a static, eternal ξ -universe there is no primordial nucleosynthesis epoch, and the Spite plateau is not the relic of an expanding-universe event. This note presents the FFGFT position, including a tentative explanation sketch for why D and ${}^4\text{He}$ abundances nonetheless agree with BBN+CMB predictions while ${}^7\text{Li}$ does not. The explanation is presented as a sketch, not a closed derivation.

1 The Problem in Numbers

The Big Bang Nucleosynthesis prediction, when fed the WMAP/Planck baryon-to-photon ratio $\eta_{10} = 6.23 \pm 0.17$, yields:

$$\text{D/H}|_{\text{BBN+CMB}} = (2.82 \pm 0.21) \times 10^{-5} \quad (\text{observation: matches}) \quad (1)$$

$$Y_p({}^4\text{He}) = 0.249 \pm 0.009 \quad (\text{observation: matches}) \quad (2)$$

$${}^7\text{Li/H}|_{\text{BBN+CMB}} = (5.24^{+0.71}_{-0.67}) \times 10^{-10} \quad (\text{observation: differs}) \quad (3)$$

The Spite-plateau measurement [1, 6] of metal-poor halo stars gives

$${}^7\text{Li/H}|_{\text{obs}} = (1.23^{+0.68}_{-0.32}) \times 10^{-10}, \quad (4)$$

a factor of 2.4–4.3 below the BBN+CMB prediction, corresponding to 4.2σ – 5.3σ [4]. Cyburt, Fields and Olive titled their 2008 update “A Bitter Pill: The Primordial Lithium Problem Worsens” – improved nuclear cross-section data *increased* the predicted ${}^7\text{Li}$ abundance.

The discrepancy is universal across stellar environments: the Spite plateau has been observed in halo field stars, globular clusters [4], dwarf spheroidals, ultra-faint galaxies, and accreted satellites [9]. It is not a feature of any one galactic environment.

2 Standard Solution Classes

Following Fields (2011), three classes have been pursued:

Astrophysical (stellar depletion). The most active current programme [9] models internal stellar mixing – atomic diffusion, parametric turbulence, rotation-induced angular momentum transport – to deplete photospheric ${}^7\text{Li}$ by a factor ~ 3 . The challenge: the Spite plateau is observationally *thin*, with little star-to-star scatter, and is found across very different stellar populations. Universal depletion across these environments is difficult to motivate physically.

Nuclear physics. Hypothetical resonances enhancing ${}^7\text{Be}$ destruction, e.g. ${}^7\text{Be}+{}^3\text{He} \rightarrow {}^{10}\text{C}^* \rightarrow p + {}^9\text{B}$ [6, 7], or revised ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ cross-sections. Recent experimental work has *tightened* the rates and *worsened* the prediction [4].

Beyond Standard Model. Decaying SUSY particles, time-varying fundamental constants [8], primordial magnetic fields. Constraint: D and ${}^4\text{He}$ must remain unperturbed, which leaves a narrow parameter window.

The fourth class. Fields (2011) closes his classification with: “*large changes to the cosmological framework used to interpret light element data.*” This category has received less attention but is logically open.

3 The FFGFT Position

FFGFT instantiates the fourth class. The cosmological framework is static and eternal, derived from the single dimensionless parameter $\xi = 4/30,000$:

- No Big Bang. The Heisenberg time-energy uncertainty argument (Doc 025) precludes a finite-duration cosmological singularity.
- No expanding spacetime. Cosmological redshift is a geometric path effect through the granular ξ -vacuum, not a Doppler signature (Doc 026).
- No primordial nucleosynthesis epoch. The Spite plateau is therefore *not* the relic of a universe-wide event 13.8 Gyr ago.

In this framework, the Spite plateau records the long-term equilibrium ${}^7\text{Li}$ abundance produced and maintained by stellar processes (Galactic cosmic-ray spallation, novae, AGB stars, core-collapse supernovae, white-dwarf mergers [9]) in a quasi-stationary state. The factor 3–4 “discrepancy” is then not a discrepancy with reality, but with a prediction that requires an event (BBN) that did not occur in this framework.

The honest follow-up question is: *why do D and ${}^4\text{He}$ match BBN+CMB so precisely if there is no BBN?*

4 Explanation Sketch: Why D and ${}^4\text{He}$ Match

What follows is an *explanation sketch*, not a closed derivation. The point is to indicate a plausible mechanism, not to claim that the mechanism is already worked

out.

Step 1 - The CMB is real, but not a relic. FFGFT accepts the CMB as a physical phenomenon, but identifies it as a manifestation of the ξ -vacuum rather than the redshifted afterglow of a hot early phase. In Doc 091, the CMB energy density is fixed by

$$\rho_{\text{CMB}} = \frac{\xi \hbar c}{L_\xi^4}, \quad L_\xi \approx 100,24 \mu\text{m}, \quad (5)$$

and the CMB temperature follows accordingly. The photon bath that carries the spectrum is real and operative *now*, not just at $z \approx 1100$.

Step 2 - The baryon-to-photon ratio. In standard BBN, light-element yields depend on the single parameter $\eta = n_B/n_\gamma$. In FFGFT, both the photon density (from ρ_{CMB} and T_{CMB}) and the cosmic baryon density (from continuous stellar processes integrated over an eternal universe) are present-day quantities. Their ratio defines an *effective* η_{eff} , which the WMAP/Planck inference correctly extracts from the CMB acoustic peaks – because the acoustic-peak structure depends only on the present-day η , not on its history.

The Λ CDM interpretation reads back from this η to a primordial η via assumed entropy conservation; FFGFT keeps the present-day η as a stationary characteristic of the ξ -universe.

Step 3 - Thermal-equilibrium yields. For light nuclides whose abundances are governed by *rapid two-body equilibrium reactions in a thermal bath*, the equilibrium abundance is a function of η and T alone, not of the history that produced this (η, T) pair. The reaction network that determines D and ^4He abundances under BBN conditions is dominated by such two-body equilibria with strong reverse reactions:



If the FFGFT ξ -vacuum sustains a $(T_{\text{CMB}}, \eta_{\text{eff}})$ that coincides with the WMAP-inferred values, the same equilibrium abundances follow. This is why D and ^4He observations agree with BBN+CMB predictions: the predictions describe a *thermal equilibrium*, and the equilibrium is realised in FFGFT for the same (T, η) by construction. The agreement is not a coincidence; it is a consistency check on the ξ -determination of T_{CMB} .

Step 4 - Why ^7Li breaks the pattern. ^7Li is the exception because its production-destruction chain has a *slow, non-equilibrium* component:

1. ^7Li at WMAP η is produced predominantly as ^7Be via ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ [6].
2. In standard BBN, ^7Be then survives the cessation of nuclear reactions and decays to ^7Li by electron capture on a timescale of ~ 53 days – after BBN has frozen out.
3. In FFGFT, no freeze-out occurs because there is no expansion-driven temperature drop. The ^7Be population sits in a stationary environment with continuous photon and electron interactions, including channels that destroy ^7Be on timescales much shorter than its laboratory half-life would suggest.

The BBN calculation assumes a clean ${}^7\text{Be} \rightarrow {}^7\text{Li}$ conversion after freeze-out, treating it as bookkeeping. In FFGFT this assumption is invalid: the stationary state allows continuous ${}^7\text{Be}$ destruction (via cosmic-ray-induced spallation, energetic photon channels in the ξ -vacuum, or stellar incorporation) that the BBN bookkeeping does not account for. The plateau-level ${}^7\text{Li}$ abundance is then the stationary attractor of the full network in a static universe, and it is naturally lower than the BBN bookkeeping value.

What this sketch does and does not claim. *Does claim:* a structural reason why an equilibrium-dominated nuclide (D, ${}^4\text{He}$) follows the standard prediction while a freeze-out-dependent nuclide (${}^7\text{Li}$) does not. *Does not claim:* a closed quantitative derivation of the Spite-plateau value from ξ . The next step would be to set up the stationary network in the ξ -vacuum and compute the attractor for ${}^7\text{Li}$. This is open work, not a finished result.

5 Summary

1. The lithium problem is real and standard: factor 3–4, 4.2σ – 5.3σ , persistent across four decades and many proposed fixes.
2. Astrophysical, nuclear and BSM solutions all face structural challenges; the cosmological-framework alternative (Fields 2011) remains open.
3. FFGFT instantiates the cosmological-framework alternative: no Big Bang, no primordial nucleosynthesis epoch, the Spite plateau is a stationary stellar-equilibrium feature.
4. The agreement of D and ${}^4\text{He}$ with BBN+CMB is not coincidental but a consistency check: equilibrium abundances depend on (T, η) alone, and FFGFT fixes both via ξ -geometry.
5. ${}^7\text{Li}$ differs because its abundance depends on a freeze-out assumption that is invalid in a static universe; the stationary attractor lies below the BBN bookkeeping value.
6. The quantitative derivation of the FFGFT ${}^7\text{Li}$ attractor is open work; this note presents the explanation as a sketch.

References

References

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FFGFT source documents: Doc 025 (*T0 Cosmology, ξ -field universe and CMB*), Doc 026 (*geometric cosmology, H_0 and the redshift mechanism*), Doc 091 (*Casimir-CMB relation, L_ξ*). All available at <https://github.com/jpascher/T0-Time-Mass-Duality> and Zenodo v1.0.13: <https://zenodo.org/records/20041529>.